

Formal Verification Report: Theorem 1.1

Black Swan Labs FCVE
ledger through EVENT-014

2026-09-19

Trust Statement. The formal result was checked using Lean 4.28.0 under the pinned project environment (Mathlib v4.28.0 pin). The theorem CLAIM-001 was kernel-checked with axiom footprint propext, Classical.choice, Quot.sound. Independent checking status is CHECKER_INCOMPATIBLE for results_eliahou_theorem_1_1 specifically (no verdict: the external checker could not process this proof's export). Computational/adversarial testing produced: COMPUTATIONALLY_SUPPORTED (diagnostic only). The remaining trust limitations are: The independent check (Gate 9) produced no verdict; The report (Gate 12) was not recorded; The proof uses Classical.choice; it is a classical proof; The repository, commit and toolchain were captured after the run (on 2026-09-19), not recorded during it; they describe the project as it is now, and the commit matches the one named in the run's own notes; The checker used for the independent check is not identified in the record; only its result is; The recorded result of EVENT-006 refers to a log or receipt that is not part of this report ("see prior receipt for full log"), so that material cannot be inspected here; No counterexample was sought against the underlying nontrivial-cycle claim; that is infeasible. This report does not claim that computational testing constitutes formal proof or that formalization alone establishes the truth of the original informal statement.

Abstract

This report documents the formal verification of Theorem 1.1 (CLAIM-001) under the Formal Claim Verification Engine. Recorded: build PASS (8033/8033 jobs; see prior receipt for full log); axiom footprint propext, Classical.choice, Quot.sound; independent check CHECKER_INCOMPATIBLE for results_eliahou_theorem_1_1 specifically (no verdict: the external checker could not process this proof's export). Computational testing is diagnostic and is not proof. Section 17 states the status the recorded evidence permits; the governance decision is recorded separately, after this report.

1 Mathematical Statement

Theorem 1.1. *Let Ω be a nontrivial cycle of T . Provided $\min \Omega > 2^40$, we have $\text{Card } \Omega = 301994a + 17087915b + 85137581c$, where a, b, c are nonnegative integers, $b > 0$, and $ac = 0$. In particular, the smallest admissible values for $\text{Card } \Omega$ are 17087915, 17389909, 17691903, and so on.*

Source location: eliahou-collatz.pdf, pages 46, 55-56.

Facts the statement relies on:

- $p_{13} = 301994$, $q_{13} = 190537$ (Table 1, $n=13$)
- $p_{15} = 17087915$, $q_{15} = 10781274$ (Table 1, $n=15$)

- $p_{16} = 85137581, q_{16} = 53715833$ (Table 1, $n=16$)
- $K(2^{39}) = p_{13} = 301994$ (p.55: the transition point $\text{tr}(13,0)$ is near 2^{39})
- $K(2^{40}) = K(2^{48}) = p_{15} = 17087915$ (p.55: the next transition point $\text{tr}(15,0)$ is near 2^{48})
- $p_{16}/q_{16} < \log_2(3) < p_{15}/q_{15} < \log_2(3+2^{-40}) < p_{13}/q_{13}$ (p.55, from Tables 1-2)

2 Definitions and Assumptions

Definitions

- $T: \mathbb{N} \rightarrow \mathbb{N}$, $T(n) = n/2$ if n even, $(3n+1)/2$ if n odd (p.45)
- trajectory: $\Omega(n) = \{n, T(n), T^2(n), \dots\}$ (p.45)
- cycle of length k : a trajectory Ω with $T^k(x)=x$ for all x in Ω , where $k = \text{Card } \Omega$ (p.45) – 'length k ' is DEFINED to equal $\text{Card } \Omega$, not merely correlated with it
- trivial cycle: $\Omega(1) = \{1,2\}$, the only cycle containing 1 under the $3x+1$ conjecture (p.45)
- $\theta = \log_2(3)$, an irrational number; $[a_0, a_1, a_2, \dots]$ its continued fraction expansion (p.50)
- p_n/q_n : the principal convergents to θ , defined recursively by $p_n = a_n p_{n-1} + p_{n-2}$, $q_n = a_n q_{n-1} + q_{n-2}$ (p.51, eq 3.2)
- Farey pair: $(p/q), (p'/q')$ with $pq' - p'q = \pm 1$ (p.49)
- $K(m)$: the smallest positive integer k such that $\log_2(3) < k/l \leq \log_2(3+1/m)$ for some positive integer l (p.48, p.52)

Remark 1. *Not recorded: assumptions.json. This input was not captured during the run, so nothing is shown in its place.*

3 Proof Outline

1. **step 1.** Corollary 2.3 (via Theorem 2.1 + definition of K, L): $\text{Card } \Omega \geq K(\min \Omega)$, $\text{Card } \Omega_1 \geq L(\min \Omega)$
2. **step 2.** Set $k = \text{Card } \Omega$, $l = \text{Card } \Omega_1$. By Theorem 2.1, $\log_2(3+M^{-1}) < k/l \leq \log_2(3+m^{-1})$; since $M \geq m > 2^{40}$, k/l lies in the open interval $(\log_2(3), \log_2(3+2^{-40}))$
3. **step 3.** That interval sits strictly between three specific convergents: $p_{16}/q_{16} < \log_2(3) < p_{15}/q_{15} < \log_2(3+2^{-40}) < p_{13}/q_{13}$
4. **step 4 case split.** k/l falls into exactly one of three regions, each handled by Lemma 3.1 (Farey-pair intermediate-fraction structure): (i) k/l in $(p_{16}/q_{16}, p_{15}/q_{15}) \Rightarrow k = p_{15}b + p_{16}c$ for some b, c in $\mathbb{N}$; (ii) $k/l = p_{15}/q_{15}$ exactly $\Rightarrow k = p_{15}b$ for some b in $\mathbb{N}$; (iii) k/l in $(p_{15}/q_{15}, p_{13}/q_{13}) \Rightarrow k = p_{13}a + p_{15}b$ for some a, b in $\mathbb{N}$

5. **step 5.** Since the three cases are mutually exclusive and each includes a nonzero p_{15} ($=17087915$) term, combining them gives $\text{Card } \Omega = 301994a + 17087915b + 85137581c$ with $b > 0$ always, and $a \cdot c = 0$ because no single case ever produces both a nonzero a AND a nonzero c simultaneously

Lemma 2.2. *Let Ω , Ω_1 be as above, and let $k = \text{Card } \Omega$. Then $\prod_{n \in \Omega_1} (3+n^{-1}) = 2^k$.*

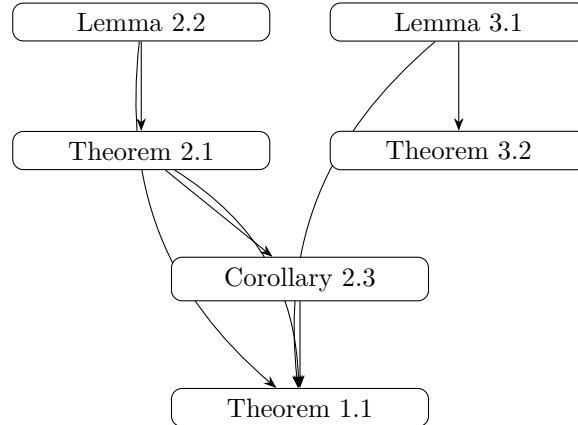
Theorem 2.1. *Let Ω be a cycle of T , and let $\Omega_1 \subset \Omega$ denote the subset of its odd elements. Then $\log_2(3+M^{-1}) < \text{Card}(\Omega)/\text{Card}(\Omega_1) \leq \log_2(3+m^{-1})$, where $M = \max \Omega$, $m = \min \Omega$. In fact, a sharper right inequality holds: $\text{Card}(\Omega)/\text{Card}(\Omega_1) \leq \log_2(3+\mu)$, where $\mu = (1/\text{Card } \Omega_1) * \sum_{n \in \Omega_1} n^{-1}$.*

Lemma 3.1. *Let $p/q < p'/q'$ be a Farey pair. Then any intermediate fraction $p/q < x/y < p'/q'$ with $y > 0$ is of the form $x/y = (ap+bp')/(aq+bq')$ where a, b are positive integers. In particular, $x \geq p+p'$ and $y \geq q+q'$.*

Corollary 2.3. *Let Ω be a cycle of T , and let $\Omega_1 \subset \Omega$ denote the subset of its odd elements. Then $\text{Card } \Omega \geq K(\min \Omega)$, $\text{Card } \Omega_1 \geq L(\min \Omega)$.*

Theorem 3.2. *Let $\theta > 0$ be irrational, θ' any number with $\theta < \theta' < [\theta] + 1$. If k, l are positive integers with $\theta < k/l \leq \theta'$, and either k or l is minimal for this property, then k/l is one of the upper convergents to θ – precisely, the largest upper convergent $\leq \theta'$.*

4 Proof Dependency Diagram



5 Formalization

Lean identifiers below are cross-references; the mathematics is in the sections above (§29).

`results_eliahou_theorem_1_1`

```

theorem results_eliahou_theorem_1_1 {L : ℕ} (c : CollatzCycle L)
  (hinj : Function.Injective c.seq)
  (hmin : (2 : ℕ) ^ 40 < c.minElem) :

```

```

∃ (a b cCoeff : ℕ), 0 < b ∧ a * cCoeff = 0 ∧
  (Finset.univ.image c.seq).card =
    301994 * a + 17087915 * b + 85137581 * cCoeff

```

6 Semantic Bridge

Informal Statement

Let Ω be a nontrivial cycle of T . Provided $\min \Omega > 2^{40}$, we have $\text{Card } \Omega = 301994a + 17087915b + 85137581c$, where a, b, c are nonnegative integers, $b > 0$, and $ac = 0$. In particular, the smallest admissible values for $\text{Card } \Omega$ are 17087915, 17389909, 17691903, and so on.

Normalized Statement

Remark 2. *Not recorded: normalized claim file. This input was not captured during the run, so nothing is shown in its place.*

Lean Statement

```

theorem results_eliahou_theorem_1_1 {L : ℕ} (c : CollatzCycle L)
  (hinj : Function.Injective c.seq)
  (hmin : (2 : ℕ) ^ 40 < c.minElem) :
  ∃ (a b cCoeff : ℕ), 0 < b ∧ a * cCoeff = 0 ∧
    (Finset.univ.image c.seq).card =
      301994 * a + 17087915 * b + 85137581 * cCoeff

```

Differences

Remark 3. *Not recorded: Gate 4 comparison. This input was not captured during the run, so nothing is shown in its place.*

Syntactic signals in the Lean statement that a reviewer must address: injectivity [`'Function.Injective'`], finite_structure [`'.card'`, `'Finset'`], strictness [`'<'`].

Findings recorded by the reviewer

- **added hypotheses** (disclosed): `hinj : Function.Injective cyc.seq` is an extra hypothesis of the Lean theorem that the paper does not state (Gate 4: 'Extra hypothesis'); it is the encoding hypothesis, see Section 31 disclosure.
- **coercions** (disclosed): The proof passes through $\mathbb{R}/\mathbb{Q}$ intermediate steps (`rat_of_real_div_lt`) that the paper does not need; this is in the proof, not in the headline statement, and Gate 4 records it as a formalization convenience because the convergents are already rational.
- **injectivity assumptions** (disclosed): `hinj : Function.Injective cyc.seq`. The paper defines cycle length k as $\text{Card } \Omega$ (p.45); Lean's `CollatzCycle L` is an L -indexed sequence that may repeat elements, so injectivity is required to identify L with $\text{Card } \Omega$.
- **classical assumptions** (disclosed): `Classical.choice` is in the axiom footprint (Gate 6). The Gate 7 note attributes it to ordinary real-number order decidability (`Real.logb`, `lt_trichotomy`), as in any Mathlib real-analysis argument.

Justification

Gate 4 records MATCH with one structural difference, injectivity: the injectivity hypothesis exactly closes the gap between Lean’s more general indexed structure and the paper’s definition, so the theorem is neither weakened nor strengthened relative to the paper. The proof also implements the paper’s own three-case Farey-pair argument. The remaining disclosed items are the R/Q intermediate steps in the proof and Classical.choice in the footprint.

Representational difference (Section 31)

What the object means informally. Omega is a set of distinct positive integers forming a cycle of the compressed Collatz map T, and the paper defines the cycle’s length k to equal Card Omega (p.45).

What Lean’s type represents. `cyc : CollatzCycle L`, an L-indexed sequence `cyc.seq` (period L, repetitions allowed); Omega is the Finset (`Finset.univ.image cyc.seq`) and Card Omega is its `.card`.

Why the encoding is valid. With `hinj : Function.Injective cyc.seq` the indexing is exactly Lean’s `ExactCollatzCycle` (injective indexing), which is what a paper ‘cycle’ inherently is; Gate 4 notes this was confirmed independently.

Where the property is preserved. Card Omega is the `.card` of the image finset in the conclusion; injectivity is what makes that image’s size equal the period L.

Whether the equality is literal or transported through an equivalence. Literal on the image finset: the conclusion states an equality of natural numbers about `(Finset.univ.image cyc.seq).card`. The step from the paper’s set Omega to the indexed sequence is the representational choice, not a transport through an equivalence.

Assumptions that make the translation valid. `hinj` (injective indexing) as an explicit theorem hypothesis, and `hmin : 2^40 < cyc.minElem` for the paper’s minimum condition.

Verdict

FAITHFUL WITH EXPLICIT REPRESENTATIONAL DIFFERENCE

Set by Wilson. Recorded context: Gate 4 recorded MATCH; Gate 7 recorded PASS – no undisclosed semantic change; Lean proves `numOdd_pos` rather than assuming it (stricter than paper).

7 Lean Environment

Lean Version	4.28.0
Mathlib Version	v4.28.0 pin

8 Formal Verification

PASS (8033/8033 jobs; see prior receipt for full log). Source of this field: ledger:EVENT-006.

9 Axiom Audit

Actual axiom footprint: `propext`, `Classical.choice`, `Quot.sound`. Source: `parsed-legacy-text:EVENT-007`. `Classical.choice` is present: the proof is classical, which is reported here and does not by itself fail the audit (§14).

10 Independent Check

`CHECKER_INCOMPATIBLE` for `results_eliahou_theorem_1_1` specifically – `lean4export` stalls on this theorem’s large proof closure (reproducible, root-caused: memoization cache reset on every call in `Export.lean`, not a resource issue). 5 of 9 paper-facing theorems (`results_eliahou_product_formula`, `results_farey_pair_bound`, `results_eliahou_product_formula_real`, `results_eliahou_sandwich`, `results_log2_three_lt_ratio`) DID pass independent check cleanly. Recording the checker and its version does not by itself establish independence (§17).

11 Computational / Adversarial Tests

Counterexample search: `COMPUTATIONALLY_SUPPORTED` – all convergents (p13,p15,p16), both Farey identities, and all three boundary inequalities confirmed. Original `MISMATCH` was a float64 precision artifact in the test script, not a defect in the claim. [diagnostic, not proof]

Computational test: `COMPUTATIONALLY_SUPPORTED` – see `EVENT-010` [diagnostic, not proof]

Neither result is formal proof (§16, §18).

What these tests do not establish

- They are not proof of the claim (Sections 16 and 18). A search that finds no counterexample is diagnostic evidence only.
- The tested domain is not recorded for these results, so this report cannot say which cases were examined.
- No counterexample was sought against the underlying claim about nontrivial Collatz cycles: doing so would require exhibiting an actual nontrivial cycle, which is computationally infeasible and open-problem-adjacent. The computational checks covered numeric and boundary structure only: the convergents p13, p15 and p16, both Farey identities, and the three boundary inequalities. (stated by Wilson)

12 Verification Limitations

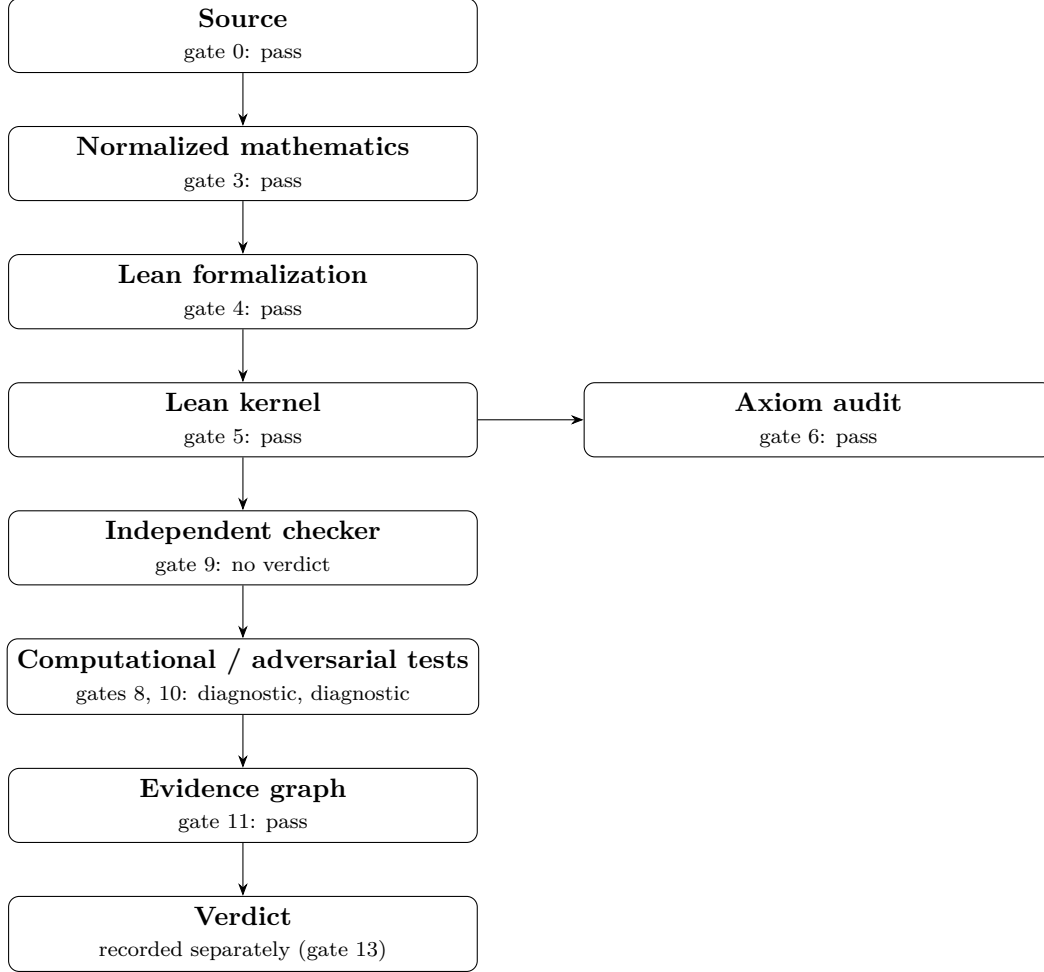
Item	Limitation
1	The independent check (Gate 9) produced no verdict. Recorded result: <code>CHECKER_INCOMPATIBLE</code> for <code>results_eliahou_theorem_1_1</code> specifically – <code>lean4export</code> stalls on this theorem’s large proof closure (reproducible, root-caused: memoization cache reset on every call in <code>Export.lean</code> , not a resource issue). 5 of 9 paper-facing theorems (<code>results_eliahou_product_formula</code> , <code>results_farey_pair_bound</code> , <code>results_eliahou_product_formula_real</code> , <code>results_eliahou_sandwich</code> , <code>results_log2_three_lt_ratio</code>) DID pass independent check cleanly.
2	The report (Gate 12) was not recorded.
3	The proof uses <code>Classical.choice</code> ; it is a classical proof.
4	The repository, commit and toolchain were captured after the run (on 2026-09-19), not recorded during it; they describe the project as it is now, and the commit matches the one named in the run’s own notes.
5	The checker used for the independent check is not identified in the record; only its result is.
6	The recorded result of EVENT-006 refers to a log or receipt that is not part of this report (“see prior receipt for full log”), so that material cannot be inspected here.
7	No counterexample was sought against the underlying claim about nontrivial Collatz cycles: doing so would require exhibiting an actual nontrivial cycle, which is computationally infeasible and open-problem-adjacent. The computational checks covered numeric and boundary structure only: the convergents p13, p15 and p16, both Farey identities, and the three boundary inequalities (stated by Wilson).
<i>Record-keeping notes (these do not change what the verification shows)</i>	
8	12 of 14 ledger events predate the input/output hash fields and carry none.
9	Events <code>EVENT-009</code> (<code>MISMATCH_FOUND</code>); <code>EVENT-010</code> , <code>EVENT-012</code> (<code>COMPUTATIONALLY_SUPPORTED</code>) used wording outside the Section 16 vocabulary; each is read as diagnostic evidence, not proof.

13 Verification Matrix

Gate	Ledger action	Event	Status	Satisfied
G0	source intake	EVENT-001	PASS	yes
G1	claim extraction	EVENT-013	PASS	yes
G2	assumption extraction	EVENT-003	PASS	yes
G3	semantic normalization	EVENT-004	PASS	yes
G4	lean formalization comparison	EVENT-005	PASS	yes
G5	lean build	EVENT-006	PASS	yes
G6	axiom audit	EVENT-007	PASS	yes
G7	semantic re audit	EVENT-008	PASS	yes
G8	adversarial computational test	EVENT-010	DIAGNOSTIC	yes
G9	independent check	EVENT-011	NO_VERDICT	no

Gate	Ledger action	Event	Status	Satisfied
G10	computational check	EVENT-012	DIAGNOSTIC	yes
G11	evidence graph	EVENT-014	PASS	yes
G12	report generation	-	MISSING	no
ORDER	canonical chain (§2)	-	PASS	yes

14 Trust / Verification Chain



15 Reproducibility Instructions

Repository URL	<code>https://github.com/tangentstorm/eliahou-collatz-bounds.git</code>
Commit	<code>db804ce6305ea99a817f067869607f8b677d895a</code> (working tree clean)
Project directory (now)	<code>/Users/lordwilson/ico-collatz/targets/eliahou-collatz-bounds</code>
Lean toolchain file	<code>leanprover/lean4:v4.28.0</code>
Lean version	4.28.0
Lake version	Lake version 5.0.0-src+7e01a1b (Lean version 4.28.0)
Mathlib revision	v4.28.0 pin (commit 8f9d9cff6bd728b17a24e163c9402775d9e6a365)
Checker	NOT RECORDED
Expected build result	PASS (8033/8033 jobs; see prior receipt for full log)
How these were obtained	Captured 2026-09-19 from the project as it is now; not recorded during the run
Matches the commit the run's notes name	yes (db804ce6)

```
git clone https://github.com/tangentstorm/eliahou-collatz-bounds.git
cd eliahou-collatz-bounds
git checkout db804ce6305ea99a817f067869607f8b677d895a
cat lean-toolchain
lake --version
lean --version
lake exe cache get && lake build
# in a Lean file importing the module: #print axioms results_eliahou_theorem_1_1
# independent-check commands were not recorded for this run
```

The repository facts were captured after the run and describe the project as it is now; they do not prove what the original run built. The build command is the one recorded in EVENT-006. Reproducibility level supported by recorded evidence: R0. This tool did not re-run the build.

16 Correction Ledger

- **TEST-001** (EVENT-009): MISMATCH_FOUND ($p_{16}/q_{16} < \log_2(3)$ boundary check failed under float64 precision)

Gates that produced no verdict (INDEPCHECK-001) are not corrections; they are listed under Verification Limitations.

17 Conclusion

The recorded evidence is summarized in the Trust Statement and the verification matrix above. Recorded failure is not a passed gate.

FORMAL PROOF (build):	PASS
AXIOM AUDIT:	PASS
SEMANTIC MATCH (Gate 4):	PASS
SEMANTIC RE-AUDIT (Gate 7):	PASS
INDEPENDENT CHECK:	NO VERDICT
COUNTEREXAMPLE SEARCH:	NO COUNTEREXAMPLE FOUND IN TESTED DOMAIN (diagnostic, not proof)
COMPUTATIONAL TEST:	NO COUNTEREXAMPLE FOUND IN TESTED DOMAIN (diagnostic, not proof)
REPRODUCIBILITY:	R0 (ceiling supported by recorded evidence)

Status the evidence permits. Promotion (PROMOTE) is not supported by the recorded evidence, because: G9 (independent check): no verdict. The reviewer may still record REPAIR, REJECT, RESEARCH or STOP. This is not a decision: the governance decision is recorded as a separate step after this report.

A Lean Source

```
theorem results_eliahou_theorem_1_1 {L : ℕ} (c : CollatzCycle L)
  (hinj : Function.Injective c.seq)
  (hmin : (2 : ℕ) ^ 40 < c.minElem) :
  ∃ (a b cCoeff : ℕ), 0 < b ∧ a * cCoeff = 0 ∧
    (Finset.univ.image c.seq).card =
      301994 * a + 17087915 * b + 85137581 * cCoeff
```

B Tests

- **EVENT-009** python3 verification/computational/tests/verify_convergents.py (initial run, native float boundary comparison): MISMATCH_FOUND (p16/q16 < log2(3) boundary check failed under float64 precision)
- **EVENT-010** python3 verification/computational/tests/verify_convergents.py (corrected: mpf 60-digit precision for boundary comparisons): COMPUTATIONALLY_SUPPORTED – all convergents (p13,p15,p16), both Farey identities, and all three boundary inequalities confirmed. Original MISMATCH was a float64 precision artifact in the test script, not a defect in the claim.
- **EVENT-012** Python/mpmath backend (per Section 19 backend abstraction; MATLAB not required): COMPUTATIONALLY_SUPPORTED – see EVENT-010

C Evidence Receipt

Field	Value
SOURCE HASH	NOT RECORDED
THEOREM ID	CLAIM-001 (Theorem 1.1)

Field	Value
CLAIM	Let Ω be a nontrivial cycle of T. Provided $\min \Omega > 2^{40}$, we have $\text{Card } \Omega = 301994a + 17087915b + 85137581c$, where a, b, c are nonnegative integers, $b > 0$, and $ac = 0$. In particular, the smallest admissible values for $\text{Card } \Omega$ are 17087915, 17389909, 17691903, and so on.
NORMALIZED CLAIM	normalized/normalized-claim.md – MATCH
LEAN STATEMENT	Gate 4: MATCH
LEAN VERSION	4.28.0
MATHLIB VERSION	v4.28.0 pin
BUILD RESULT	PASS (8033/8033 jobs; see prior receipt for full log)
AXIOM FOOTPRINT	propext, Classical.choice, Quot.sound
INDEPENDENT CHECK	CHECKER_INCOMPATIBLE for results_eliahou_theorem_1_1 specifically – lean4export stalls on this theorem’s large proof closure (reproducible, root-caused: memoization cache reset on every call in Export.lean, not a resource issue). 5 of 9 paper-facing theorems (results_eliahou_product_formula, results_farey_pair_bound, results_eliahou_product_formula_real, results_eliahou_sandwich, results_log2_three_lt_ratio) DID pass independent check cleanly.
COMPUTATIONAL TEST	COMPUTATIONALLY_SUPPORTED – see EVENT-010 [diagnostic, not proof]
COUNTEREXAMPLE SEARCH	COMPUTATIONALLY_SUPPORTED – all convergents (p13,p15,p16), both Farey identities, and all three boundary inequalities confirmed. Original MISMATCH was a float64 precision artifact in the test script, not a defect in the claim. [diagnostic, not proof]
SEMANTIC MATCH	Gate 4: MATCH; Gate 7: PASS – no undisclosed semantic change; Lean proves numOdd_pos rather than assuming it (stricter than paper)
REPRODUCIBILITY LEVEL	R0 (ceiling supported by recorded evidence; NOT re-run by this tool; R5 needs full-package rebuild, not assessed)
GRAPH HASH	v3 8558c9b7f7b0e9aeb60b0157f026864f77d1440d54954caa810723ae9a62f292 (covers 14 events through EVENT-014)
COST	14 ledger events; LEAN_BUILD_TIME=NOT RECORDED; COMPUTATIONAL_TEST_TIME=NOT RECORDED; CHECKER_TIME=NOT RECORDED; HUMAN_TIME, MODEL_CALL_COUNT, MODEL_COST, LATEX_BUILD_TIME=NOT RECORDED (no cost ledger, §43)